THE MASTERMIND COACHING CLASSES
CBSE / RBSE • NURSERY – CLASS XII • JAIPUR
Road No. 5, VKIA, Jaipur — 9887258679

ANSWER KEY & MARKING SCHEME SET A

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(d)** Robert Hooke	compartments in cork
2	**(d)** unicellular	e.g. amoeba, yeast
3	**(c)** allows only some substances to pass	controls entry and exit
4	**(b)** higher to lower concentration	concentration gradient
5	**(d)** plasmolysis	membrane pulls away
6	**(b)** nucleoid	no nuclear membrane
7	**(b)** ATP	adenosine triphosphate
8	**(d)** chloroplasts	photosynthesis
9	**(d)** cell sap	stores water and wastes
10	**(c)** totipotency	Haberlandt

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	All living things are made of one or more cells; the cell is the basic unit of structure and function; new cells arise from pre-existing cells (2).	2
12	Robert Hooke (1); box-like compartments in a thin slice of cork (1).	1+1
13	Unicellular: one cell, e.g. amoeba; multicellular: many cells, e.g. humans (1+1).	1+1
14	Total = 10 × 40 = 400X (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Made of lipids and proteins (1); molecules can move sideways, flip and rotate (1); arranged like tiles in a mosaic (1).	1+1+1
16	Place equal pieces in water and in 20% salt solution (1); weigh before and after (1); the piece in water gains weight and in salt solution loses weight (1).	1+1+1
17	The raisin in water swells as water enters by osmosis (1.5); in sugar solution it shrinks as water leaves (1.5).	1½+1½
	OR: Cell wall present only in plant cells; plastids only in plants; large central vacuole in plants (1+1+1).	1+1+1
18	Shrinkage of the cell contents away from the cell wall in concentrated solution (1.5); put 20% sugar solution on the peel and observe under the microscope (1.5).	1½+1½
	OR: Plants cannot move and need rigid support to stay upright (1.5); animal cells need flexibility for movement and changing shape (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) All living things consist of cells; the cell is the unit of structure and function; cells arise from cells (2). (ii) Schleiden, Schwann and Virchow (1). (iii) All life processes occur in cells (2).	2+1+2
	OR: (i) Lenses magnify an illuminated specimen (2). (ii) Resolution: clarity; contrast: difference in brightness (2). (iii) 1000X (1).	2+2+1
20	(i) Water: gains weight; salt: shrinks (1). (ii) Water moves from dilute to concentrated solution through the membrane (2). (iii) For a fair comparison (1). (iv) No net change (1).	1+2+1+1
	OR: (i) Fluid mosaic of lipids and proteins (2). (ii) Selectively permeable (1). (iii) Osmosis is the diffusion of water across a membrane; diffusion needs no membrane (2).	2+1+2

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.

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ANSWER KEY & MARKING SCHEME SET B

Chapter 9: **Cell: The Building Block of Life** | Class 9 Science | Maximum Marks: 40

SECTION — A (Objective Type) — Answers — 10 × 1 = 10

Q.	Correct Option	Brief Reasoning
1	**(a)** Theodor Schwann	animal cells
2	**(b)** 400X	10 × 40
3	**(d)** osmosis	dilute to concentrated
4	**(a)** the cell membrane	lipids and proteins
5	**(b)** nucleus	contains DNA
6	**(a)** human red blood cell	more space for haemoglobin
7	**(a)** Golgi apparatus	modifies and sorts
8	**(c)** chromoplasts	yellow, orange, red
9	**(b)** mitosis	growth and repair
10	**(c)** cell culture	tissue culture

SECTION — B (Very Short Answer) — Answers — 4 × 2 = 8

Q.	Expected Answer / Value Points	Marks
11	It encloses the cell and controls the entry and exit of substances (2).	2
12	The diffusion of water across a selectively permeable membrane from a dilute to a concentrated solution (2).	2
13	Diffusion: movement of particles from high to low concentration without a membrane; osmosis: diffusion of water across a selectively permeable membrane (1+1).	1+1
14	The rigid cell wall resists the pressure from absorbed water (2).	2

SECTION — C (Short Answer) — Answers — 4 × 3 = 12

Q.	Expected Answer / Value Points	Marks
15	Nucleus: absent vs present with membrane; organelles: no membrane-bound organelles vs present; size: smaller vs larger (1+1+1).	1+1+1
16	Diameter = 25 × 200 µm = 5000 µm (2) = 5 mm (1).	2+1
17	Transport network inside the cell (1); RER has ribosomes and makes proteins; SER has no ribosomes and makes lipids (2).	1+2
	OR: Receives, modifies, sorts and packages proteins and lipids into vesicles for transport and secretion (3).	3
18	On damage to the cell, lysosomes burst and their enzymes digest the cell itself (1.5); they also clean the cell of waste (1.5).	1½+1½
	OR: Both have their own DNA and ribosomes (1.5); both can make some of their own proteins and divide on their own (1.5).	1½+1½

SECTION — D (Long Answer) — Answers — 2 × 5 = 10

Q.	Expected Answer / Value Points	Marks
19	(i) Into X and out of Y (1). (ii) The solution around Y (1). (iii) The rigid cell wall resists the pressure (1). (iv) Plasmolysis (2).	1+1+1+2
	OR: (i) Labelled diagram showing cell wall, membrane, cytoplasm, nucleus, vacuole, chloroplast, mitochondria (3). (ii) Cell wall and plastids (1). (iii) Chloroplast (1).	3+1+1
20	(i) Controls cell activities and stores DNA (1). (ii) Chromatin is entangled DNA threads; condenses into chromosomes; genes are functional segments of DNA (3). (iii) They lack a nucleus (1).	1+3+1
	OR: (i) Energy; transport network; sorting and packaging; digestion; protein synthesis (3). (ii) Mitochondria and plastids (1). (iii) Ribosomes (½). (iv) Lysosomes (½).	3+1+½+½

Evaluator's note: Award step marks as shown; accept any correct alternative method. In internal-choice questions, evaluate only the option attempted. Deduct up to ½ mark for missing units or figures where required.